

SVM PMSM Drive With Low Resolution Hall-Effect Sensors

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Abstract—In this paper, a back-electromotive force-based method is used in conjunction with three very low-cost integrated circuits based on the Hall-effect to estimate the rotor position of the direct-drive permanent-magnet synchronous motors (PMSMs) to be used for traction purpose of an electric wheelchair. A speed estimator based on a rotor frame machine model has been implemented for the PMSM drives, and then the rotor position estimation is achieved by means of a discrete integration of the estimated speed. The three Hall effect sensors are used to detect the rotor initial position as well to reset the error on the rotor position estimation every 60 electrical degrees. In the specific application, support vector machine technique has been implemented together with the field oriented control. Simulations and experimental results are shown in the paper.

Index Terms—Encoderless, low rotational speed, permanent-magnet (PM) motor drive.

I. INTRODUCTION

THE permanent-magnet synchronous motor (PMSM) is attracting attention in recent years and it is widely recognized to be a very suitable candidate for hybrid and electric vehicles' applications. Maintenance free operation, robustness against environment, high efficiency, high power density, and high controllability are some of the PMSM characteristics responsible for its wide utilization in traction applications such as electric vehicles (EVs) and hybrid EVs (HEVs). In PMSM drives, position sensors are required to time the sinusoidal current waveforms with the rotor position; suitable waveforms of phase currents and low motor torque ripple ask for high resolution sensors such as either optical encoders or electromagnetic resolvers. However, these sensors are expensive and their coupling very often requires special constructions for the machine, such as a second shaft end; besides, position sensor misalignment can often occur during operation causing unexpected current overloads. In direct drive PMSMs which are widely used for traction applications it is required a periodic monitoring of the sensor alignment and, if necessary, the software correction of the reference position angle.

The requirement of removal of unreliable and expensive position sensors has led academic and industrial researchers to investigate and to propose several methods for the sensorless control of electrical machines. Within the last decade, significant

improvements have been made in the field of sensorless control of PMSMs; major methods for sensorless position estimation can be classified in two main groups, the first group collects methods which use back-electromotive-force (EMF) estimation with fundamental excitation, whereas the second group is related to spatial saliency image tracking methods which use excitation in addition to the fundamental [1]. The saliency tracking methods are appropriate for zero and near zero-speed operation, however machines with suitable saliency are required for correctly estimate the rotor position; back-EMF-based methods are easy to be implemented for symmetric PMSMs, however the failure of these methods at low rotational speed preclude their acceptance in direct drive PMSMs for traction applications.

In this paper a back-EMF-based method is used in conjunction with very low resolution sensors [2]–[5] to estimate the rotor position of direct-drive PMSMs to be used for traction purpose of an electric wheelchair. A speed estimator based on rotor frame machine model (SERF) has been implemented for the PMSM drives, and then the rotor position estimation is achieved by means of a discrete integration of the estimated speed. During the manufacturing process three very low-cost integrated circuits based on the Hall-effect are suitable positioned in the machine stator in order to provide 60 electrical degrees resolution in rotor position sensing, thus the error on the rotor position estimation is reset every time the rotor's magnetic axes enters a new 60° sector univocally identified by means of the three Hall-effect sensors' binary code. The Hall-effect sensors do not require any tuning process and are also used to detect the initial position of the machine rotor, in this case the rotor's magnetic axes is positioned at the half of the 60° electrical sector identified by means of the sensors' binary code. As a consequence the drive is able to start in sinusoidal operation at a torque which is in the range of 86.6%–100% of the maximum torque, in fact the maximum error in the initial position detection is 30 electrical degrees and it is reset to zero at the very first transition of any of the Hall-effect sensors.

II. WHEELCHAIR PMSM DIRECT DRIVES

To date, the drive train of electric wheelchairs is still arranged by means of old-fashion chopper-fed dc motor drives which include a gearbox for matching the low speed at which the wheel must rotate with the relatively higher rating speed of the dc motor. The use of a gearbox negatively influences mass, cost and reliability of the drive train, as well as constrains the overall arrangement of the wheelchair chassis. In addition to that, due to the use of a gearbox the overall efficiency of the today's wheelchair motor drives hardly exceeds 60%, and this leads to waste a significant amount of the battery-stored energy being available on board for the propulsion. To the purpose of improving the

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Fig. 1. Prototype of the electric wheelchair.

wheelchair performance, the use of a direct drive motor is desirable in order to eliminate the gearbox, enhance the wheelchair range of autonomy and make the wheelchair drive train cheaper, quieter and more reliable. Conventional electric motors based on the radial-field concept generally have poor torque-to-mass ratio for wheel direct drive applications and thereby would result in a relatively bulky arrangement of the drive.

The advent of high-field permanent magnet materials has opened up opportunities for electrical machine topologies with substantially improved performance, typically in terms of higher torque-to-mass ratio and improved efficiency. Materials such as Nd-Fe-B, for example, enable adequate flux densities to be produced, even with quite large air gaps, thus improving the prospects for designs of direct drive wheel motors. In the last decade, slotless axial-flux permanent magnet (AFPM) machines proved to be particularly suitable for low-speed high-torque applications, as in comparison with other machine topologies they allow direct drive motor design which results in higher lightness and compactness of the machine [6].

Previous project activities were finalized to design and construction of two twin AFPM wheel-motor drives for the replacement of the standard dc motor drives a commercially available wheelchair chassis was early equipped with. The prototype of electric wheelchair having direct-drive motors being housed within the wheel rims and designed to achieve performance as required for such a particular 24-V battery-operated vehicle application is shown in Fig. 1 [7].

Due to their modes of operation, electric wheelchairs have motion requirements which somewhat differ from those ones of others road vehicles, namely low rating speed (e.g. maximum speed of $10 \div 15$ km/h on a flat road), smooth driving sensation and maneuverability at very low speed for indoor movement, as well extremely higher peak torque in the form of a pulse of short duration (e.g., a single-shot peak torque of $5 \div 6$ times the rated torque along few decades of seconds). The high reliability required for such electric drives as well the location of the AFPM machines (i.e., in each wheel rim) discourages the use of either optical encoders or electromagnetic resolvers, thus in this

TABLE I
DESIGN CHARACTERISTICS OF DIRECT-DRIVE MOTORS

Number of poles	16
Rating power [W]	200
Nominal speed [rev/min]	136
Continuous torque [N-m]	14
Peak torque [N-m]	84
Phase peak emf @ nominal speed [V]	8.7
Phase rms current @ rated torque [A]	11
Efficiency @ rating power	86%
Number of phases	3
Phase resistance @ working temperature [mΩ]	90.5
Phase inductance [μH]	56
Machine overall axial length [mm]	55
Total mass of active materials [kg]	6.9

paper a back-EMF-based method is used, in conjunction with three Hall-effect sensors to be mounted in the machine stator, to estimate the rotor position of the AFPM motors as well to accomplish a near-zero speed control for the sinusoidal drives. In order to perform the required indoor high maneuverability the minimum continuous rotational speed of the direct-drive motors must be at least 3 rev/min. Table I summarizes the characteristics of the constructed wheelchair direct-drive motor prototypes.

III. SPEED ESTIMATOR BASED ON ROTOR FRAME MACHINE MODEL (SERF)

The speed estimator based on rotor frame machine model (SERF) is achieved by using the mathematical expressions in (1) which represent the d - q axis model of the PMSM

$$\begin{aligned} V_q &= R_s \cdot i_q + p\lambda_q + \omega_e \cdot \lambda_d \\ V_d &= R_s \cdot i_d + p\lambda_d - \omega_e \cdot \lambda_q \end{aligned} \quad (1)$$

where $\lambda_d = L_d \cdot i_d + \lambda_{af}$, $\lambda_q = L_q \cdot i_q$, L_d and L_q are the machine phase inductances on respectively d and q axes, R_s is the machine phase resistance on d - q axes, λ_{af} is the total rotor flux and ω_e is the electrical speed.

The rotational speed is estimated on the basis of the knowledge of currents and voltages on d - q axis, both currents and voltages can be directly measured in the three-phase reference and then transformed on d - q axis by means of Park and Stanley matrixes. However, while current sensors are already present in the electric drive and required for the torque control loop, inverter output line-to-line voltages should be appropriately measured by means of sensors in order to calculate d - q voltages used in (1).

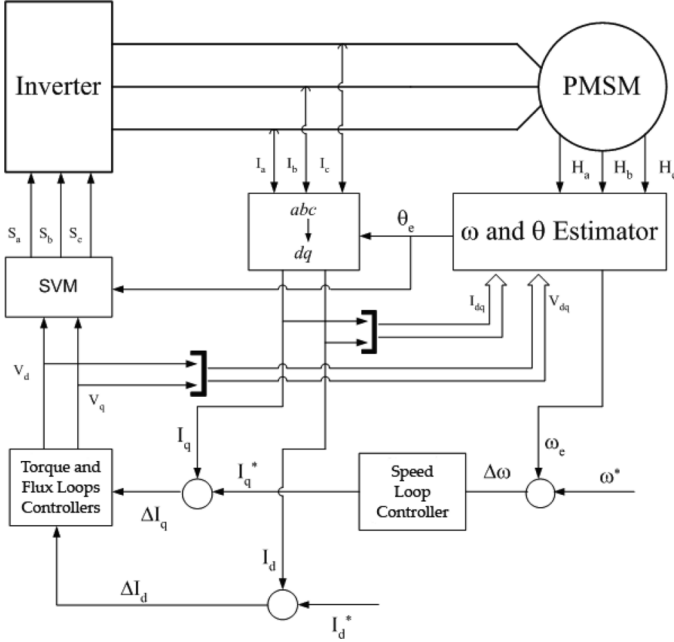


Fig. 2. Block diagram of PMSM drive with proposed SERF algorithm.

Voltage sensors can be saved when the SERF algorithm required values for d - q voltages can be directly achieved from d - q reference voltages calculated in the torque loop controller, in this case a suitable term taking in account the inverter voltage drop should be considered in the model of the speed estimator. As a consequence rotor electrical speed ω_e and rotor electrical position θ_e are calculated by means of, respectively, (2) and (3) where the term $R_{inv} \cdot i_{q(i)}$ takes fairly in account the voltage drop in the power inverter, T_s is the sampling time used for SERF algorithm, and $L_s = L_d = L_q$. The control algorithm as well the speed and position estimator for each wheelchair's motor drive has been implemented on 21992 DSP from Analog Devices, where the sampling time has been set to 400 μ s. Three Hall-effect sensors are properly positioned in the machine stator in order to provide 60 electrical degree resolution in rotor position sensing, then the SERF error of the rotor position estimation is reset every time the rotor's magnetic axes enters a new 60° sector univocally identified by means of the three Hall-effect sensors' binary code

$$\omega_{e(i)} = \left[V_{q(i)} - R_{inv} \cdot i_{q(i)} - R_s \cdot i_{q(i)} - L_s \cdot \left(\frac{i_{q(i)} - i_{q(i-1)}}{T_s} \right) \right] \cdot \frac{1}{\lambda_{af} + L_s \cdot i_{d(i)}} \quad (2)$$

$$\vartheta_{e(i)} = \vartheta_{e(i-1)} + \frac{\omega_{e(i)} + \omega_{e(i-1)}}{2} \cdot T_s. \quad (3)$$

Block diagram for each one of the wheelchair AFPM drives is shown in Fig. 2; support vector machine (SVM) logic block and speed estimator block receive the same d - q axis voltages as inputs, SERF algorithm provides both electrical speed and position to the drive control section, in particular the position information is directly used by SVM block in order to calculate the inverter switching pattern.

IV. SVM AND HALL EFFECT POSITION SENSORS

The investigated traction drive has the peculiarity of low-voltage power supply and it is characterized by high

TABLE II
SSVS CONFIGURATIONS FOR THREE-LEGS VSI

Config.	SSV	V_{ab}	V_{bc}	V_{ca}
0 0 0	V7	0	0	0
0 0 1	V5	0	-Vdc	Vdc
0 1 0	V3	-Vdc	Vdc	0
0 1 1	V4	-Vdc	0	Vdc
1 0 0	V1	Vdc	0	-Vdc
1 0 1	V6	Vdc	-Vdc	0
1 1 0	V2	0	Vdc	-Vdc
1 1 1	V8	0	0	0

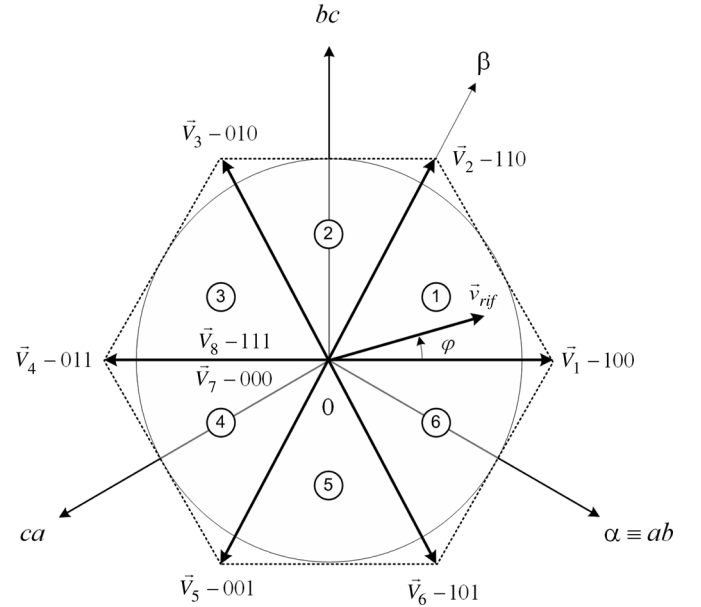


Fig. 3. Vector representation of three-inverter.

over-torque—and then high over current—mode of operation, thus suitable solutions have to be adopted to reduce voltage drops and to supply the drives with the potential maximum voltage. To this purpose the axial flux permanent magnet (AFPM) motors are fed by Mosfet inverters having low voltage drops and operating at 30 kHz switching frequency. As well, SVM technique has been implemented together with the field oriented control with the purpose to extend as much as possible the linear modulation range. It should be noticed that SVM techniques, which are based on vector representation of the inverter, divide the plane in six sectors identified by a binary code in the same way the Hall effect sensors provide 60 electrical degree resolution in the rotor position sensing; thus, sector identification from Hall sensors and estimated position from SERF algorithm can be promptly used in SVM logic saving mathematical operations and total execution time.

Recent development of microprocessors has led SVM technique to be one of the most important methods for generating

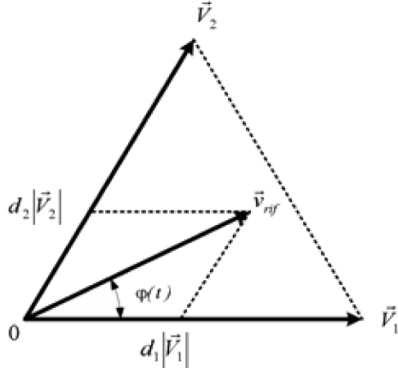


Fig. 4. Duty-cycle calculation for sector 1.

switching patterns for three-phase converters [8]. Differently from Sinusoidal PWM, the SVM considers the inverter as a unit. The vector technique accomplishes a better utilization of the entire conversion system and it's good for vector control of three-phase electrical machines. Digital control allows flexibility in controlling algorithms, and modulation schemes can be very easily implemented; furthermore, by selecting different criteria in generating the switching pattern, the power system performance and efficiency can be significantly affected [9].

SVM techniques are based on the vector representation of the inverter; the state of each leg is represented by the value of the variable S_x ($S_x = 1$ if top switch is on and the bottom one is off, $S_x = 0$ vice-versa), thus three-legs inverters can assume eight different configurations as shown in Table II. Through mathematical transformations, symmetric three-phase systems can be changed in equivalent two-phase systems: in the $\alpha\beta$ plane a rotating vector is obtained. For each configuration of the inverter, we can achieve 8 different vectors (six non-zero vectors and two zero vectors) called switching state vectors (SSVs) that divide the $\alpha\beta$ plane in six sectors. As an example considering topology 100, from Table II, the corresponding SSV is V_1 and the line voltages V_{ab} , V_{bc} , V_{ca} are, respectively, equal to V_{dc} , 0, $-V_{dc}$. Fig. 3 shows the $\alpha\beta$ plane vector representation of three-legs inverter.

Modeling of SVM technique and corresponding SSVs has been achieved in Matlab-Simulink and then implemented on 16 b-fixed point DSP platform. Inputs for SVM block are the angular information (θ_e) and the desired output voltage in $d-q$ rotating two-phase frame (V_d , V_q), the switching pattern is the output. In SVM block model information on angle θ_e and the 60° sector, in which the rotor's magnetic axes is positioned, allow equations written for sector I to be used for further five sectors in the same form; mentioned equations, used to find out duty cycle values, are shown in Fig. 4 in relation to sector I. Each SSV is applied for a period resulting from duty-cycle calculations, where dead time effects are included and maximum and minimum conduction period values are limited on the basis of actual switches characteristics.

Compensation of dead time effects is achieved as suggested in [10] in order to reduce zero current clamping. As the phase current lies inside the zero current clamping region the compensation voltages are $\Delta V_q = (1/T_s)[i_a \cdot L_s + ((2/3)V_{dc} - \lambda_{af}\omega_e \cdot \cos \vartheta_e)T_d]$ and $\Delta V_d = 0$ when the current of the phase a is negative, and $\Delta V_q = (1/T_s)[i_a \cdot L_s - (\lambda_{af}\omega_e \cdot \cos \vartheta_e)T_d]$

$$V_{ref} = \sqrt{V_d^2 + V_q^2} = d_1 V_1 + d_2 V_2; \quad V_1 = V_2 = V_{dc} \cdot 2/3; \quad V_{ll-pk} = \sqrt{3} \cdot V_{ref};$$

$$d_1 = M \sin[60 - \theta_e - 60^\circ \cdot (S-1)]; \quad d_2 = M \sin[\theta_e - 60^\circ \cdot (S-1)]; \quad d_0 = 1 - d_1 - d_2$$

with $d_x = t_x/t_s$ duty-cycle and $M = V_{ll-pk}/V_{dc}$ modulation index; where: $S = 60^\circ$ sector number; t_x = turn-on time of SSVs; t_s = switching period; V_{ll-pk} = line-to-line voltage peak value; V_{dc} = dc link voltage.

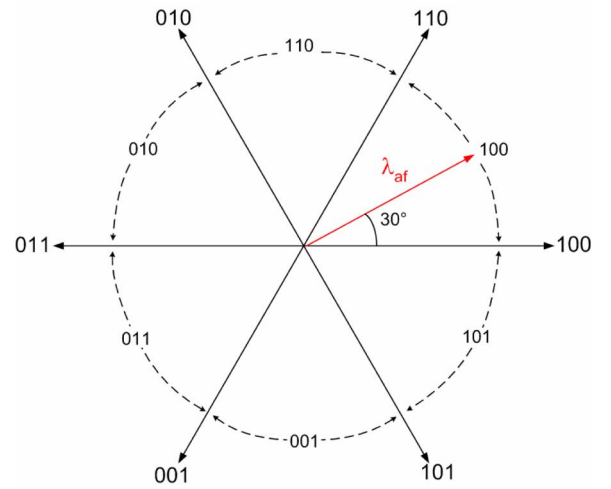


Fig. 5. Rotor's magnetic axes position at very first startup.

and $\Delta V_d = 0$ when the current of the phase a is positive; similar expressions are used for the currents of phase b and c .

At the very first start, absolute position is obtained from the three Hall-effect sensors being faced to one motor pole pair. As shown in Fig. 5, the rotor's magnetic axes position, with respect to the hexagon locus referred to the six possible combinations of the three Hall effect sensors, is assumed at the half of the 60° electrical sector identified by means of the sensors' binary code. As a consequence the drive is able to start in sinusoidal operation at a torque which is in the range of 86.6%–100% of the maximum torque, in fact the maximum error in the initial position detection is 30 electrical degrees and it is reset to zero at the very first transition of any of the Hall-effect sensors. Being the sampling time in the control algorithm set at $400 \mu s$, the new values of $\omega_{e(i)}$ and $\theta_{e(i)}$ are achieved every $400 \mu s$; as a result, the speed resolution of SERF algorithm for the proposed application (16 poles PM machine) at the minimum continuous rotational speed (i.e., 3 rev/min) is about $12\times$ higher with respect to the maximum resolution achievable from an encoder with 1024 pulses per revolution.

V. MODELING AND EXPERIMENTAL RESULTS

Simulink model for the SVM PMSM drive has been arranged. In the realized models sampling and A/D conversion of input signals as well all the other functions are accomplished simulating same operational and calculus characteristics of

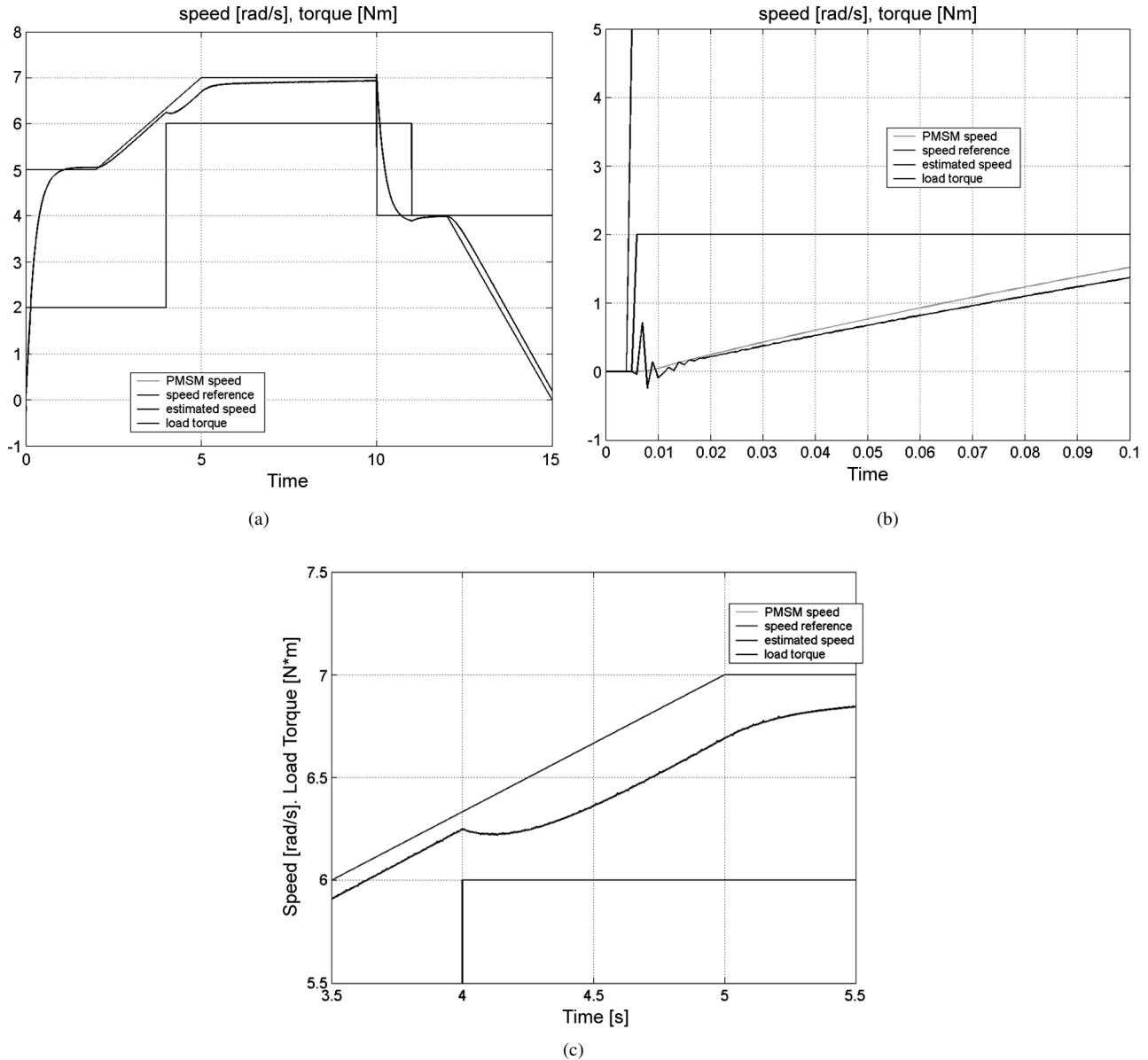
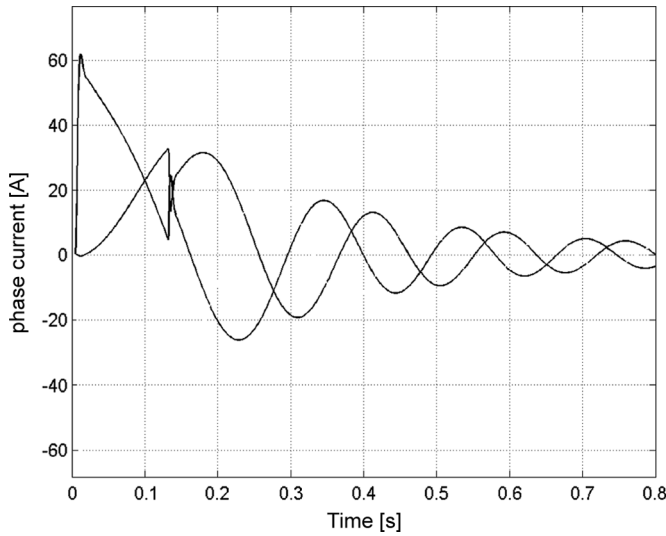


Fig. 6. (a) Simulation results: speed estimation at speed reference and load torque variations. (b) Simulation results: zoom at startup. (c) Simulation results: zoom at load variation.

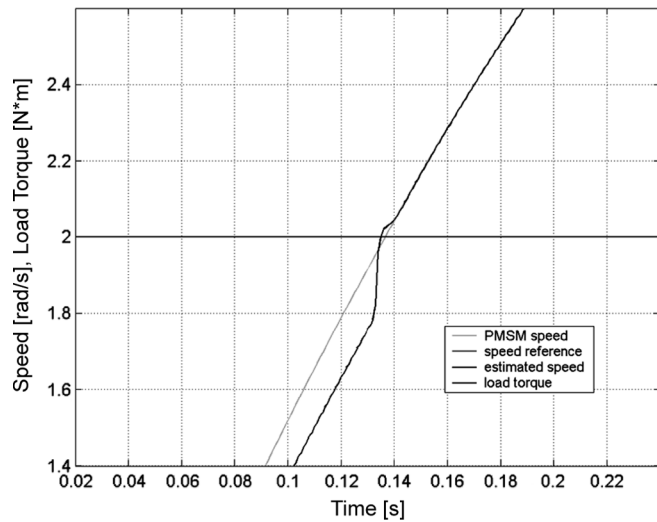
16 b-fixed point DSP. Suitable routines have been adopted in order to achieve 32 b-floating point calculus performances. In the model are present both the SERF algorithm and a 1024 pulses encoder, used in order to compare speed estimation with the measured machine rotor speed; furthermore, switching model of inverter fed AFPM drive has been accomplished by using real components.

In order to investigate the reliability of the proposed SERF algorithm for the speed estimation in PMSM drives, a sensibility analysis has been accomplished by simulations both by varying the PMSM parameters (i.e. phase resistance and inductance) and by adding to the current measures some noise at various frequencies. The motor phase inductance has been reduced by 50% whereas the stator resistance has been changed from the rated value (90 m Ω at 25 $^{\circ}$ C) to 133.35 m Ω , correspondent to the winding operating temperature of 150 $^{\circ}$ C. The frequency of the noise signals introduced in the acquisition process has been changed in the range 1.25–2.5 kHz (i.e., respectively, the half

and the same value of the sampling frequency in the control algorithm), whereas the amplitude of the noise added to the measure signals is respectively 1% (0.24 V) of dc link voltage and 1% (1 A) of the current full scale measure. It has been demonstrated that the SERF algorithm is mainly sensitive to PMSM phase resistance variation, whereas the phase inductance value hardly affects the speed and position estimation and the noise can be effectively filtered by means of Butterworth II order digital filters. As a result, either the measurement or a roughly estimation of stator resistance value must be provided to achieve proper results from (2) and (3). The error of 48% on the stator phase resistance value causes the error in the estimated speed and position of almost 30% in the most critical operating condition, which occurs when the machine rotates at the minimum continuous speed (i.e., 3 rev/min), whereas the error of 5% is responsible for 3.5% error in the estimation algorithm which drops to 1.1% when the rotational speed is 1 rad/s. It is evident the rotational speed per cent error significantly decreases at higher



(a)

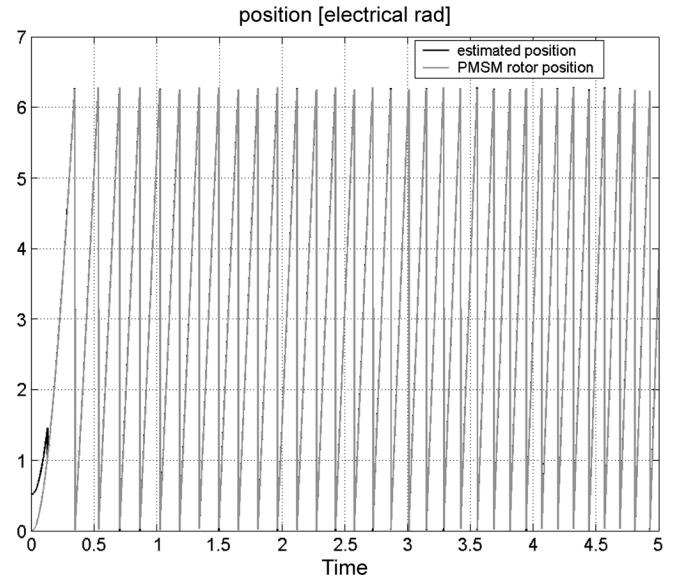


(b)

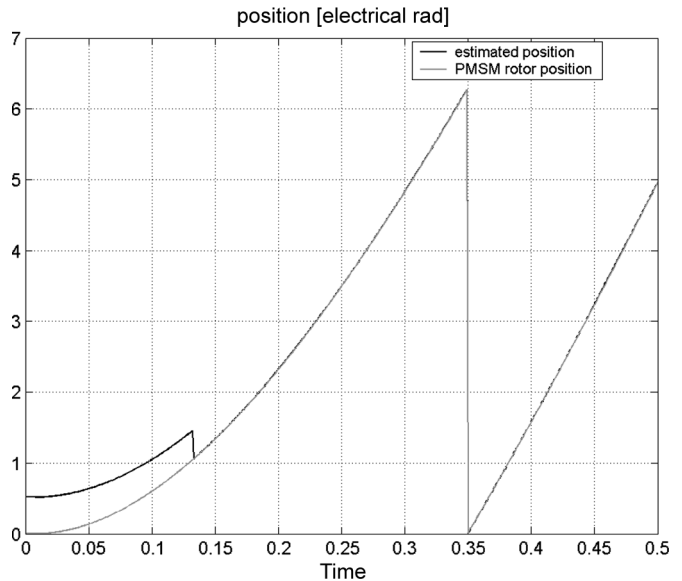
Fig. 7. (a) Simulation results: PMSM phase currents at startup. (b) Simulation results: motor speed at first reset.

speed values, which are common to outdoor modes of operation. By sensing the operating temperature of the stator winding it is possible to achieve the desired accuracy for the estimation of the value of the phase resistance; however, even in absence of a temperature sensor, the knowledge of the phase current can be used to estimate the winding resistance with approximation which could be acceptable [11].

Fig. 6 shows a simulation achieved by means of the realized Simulink model of the whole wheelchair; both speed reference and load torque is changed and in any case the speed estimation from SERF algorithm is almost coincident with motor measured speed from a 1024 pulses encoder. The simulation test starts with the maximum error in the rotor position (i.e., 30 electrical degrees), the step variation 0–5 rad/s for the motor speed reference and the step variation 0–2 Nm for the load torque are commanded; then, after the motor has reached the new steady state condition, a ramp variation (5–7 rad/s in 3 s) for the speed reference is applied, at 2/3 of the speed reference ramp the load



(a)



(b)

Fig. 8. (a) Simulation results: position estimation. (b) Simulation results: zoom at startup.

torque is changed from 2 to 6 Nm. At the second 10 of the simulation the commanded speed is decreased to 4 rad/s and 1 s later the applied torque changes from 6 to 4 Nm, then the speed reference is brought to 0 rad/s in the next 3 s. Along the entire simulation the estimated speed and the actual measured speed are tight together thus confirming the accuracy of the estimation algorithm. The considered application does not require high control bandwidth; as a consequence, the low control bandwidth saves the filtering of speed and position and improves the system stability. The same estimation and control algorithms could be applied to drives requiring better performances, however in this case speed and position filtering should be accepted. Fig. 6(b) shows a zoomed window at very first startup of the wheelchair, where the estimated speed has some small oscillations, mainly due to the effect of the applied torque step when

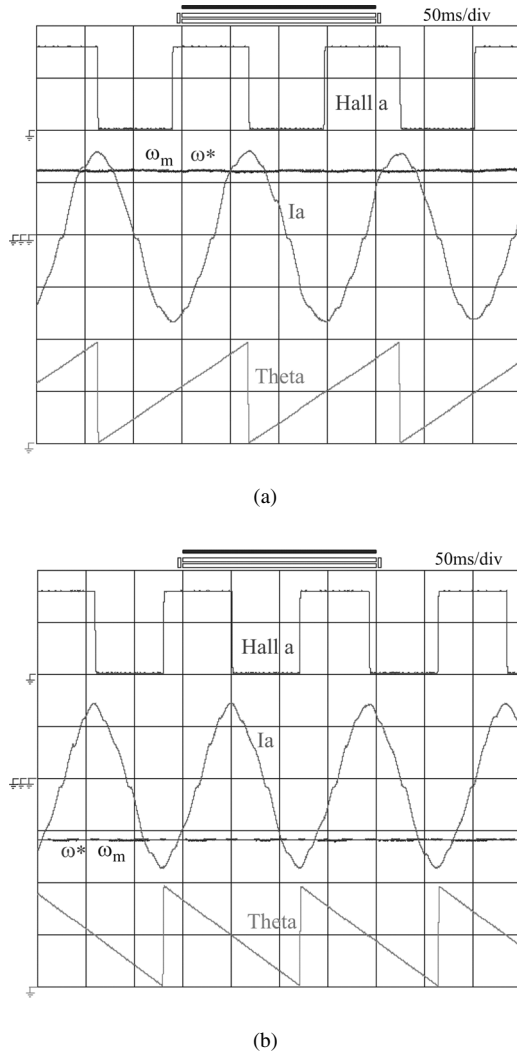


Fig. 9. Experimental results at 5 rad/s: actual speed ($4 \text{ rad} \cdot \text{s}^{-1} / \text{div}$), estimated speed ($4 \text{ rad} \cdot \text{s}^{-1} / \text{div}$), phase current ($10 \text{ A} / \text{div}$), estimated position ($\pi \text{ rad} / \text{div}$), Hall effect sensor; positive rotational speed.

the rotational speed of the machine is still zero, that however do not affect the start-up of the machine. The speed oscillations are experienced only at starting from zero and they can be mainly related to the sudden increase of V_d and V_q due to the proportional part of torque and flux loops controllers; further investigation showed that speed oscillations at starting are significantly reduced when SERF algorithm $d-q$ voltages are achieved by means of measured AFPM motor line-to-line voltages and transformed by means of the estimated position. Incidentally, all the experimental testing activity has been accomplished by using V_d and V_q values at the output of torque and flux loops. Zoomed window in Fig. 6(c) emphasizes good results for speed estimation even at load variation.

The phase currents of the PMSM are shown in Fig. 7(a) for the first 0.8 s of the simulation; at start-up one of the phase current is equal to zero because rotor's magnetic axes position is assumed at the half of the 60° electrical sector identified by means of the Hall sensors' binary code, at 0.13 s the rotor's magnetic axes enters the first new 60° sector and the initial error on the rotor position estimation is reset. From this point on the error in speed estimation is significantly reduced as shown in Fig. 7(b)

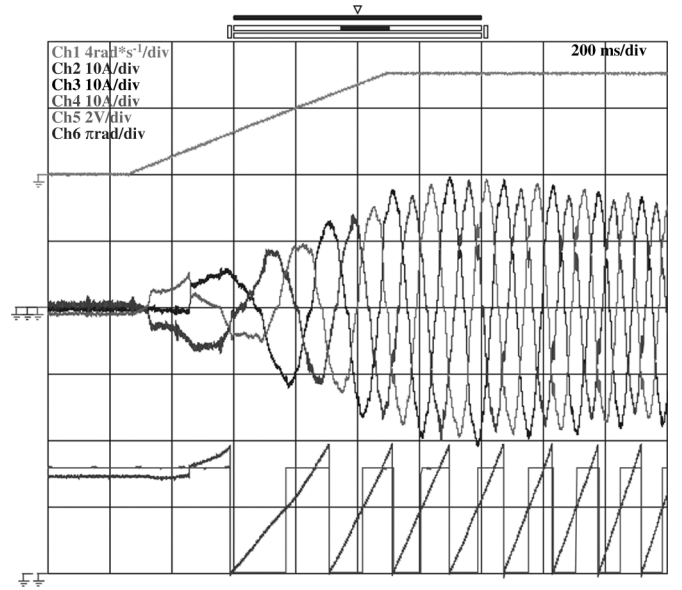


Fig. 10. Experimental results 0–6 rad/s: currents of stator phase A, B, C, reference speed, rotor estimated position, and Hall effect sensor.

and the phase currents are correctly timed with the respective EMF waveforms.

Fig. 8 shows the comparison between the estimated position and the actual position of the PMSM rotor, these simulation results confirm the good estimation achieved with SERF algorithm and the correct assumption for the start-up of the machine. It is clearly shown the assumed 30° error in the initial rotor position, which is then reset at 0.13 s.

First experimental tests, on the prototypal AFPM motor drive using the SERF algorithm in the control loop, have been devoted to prove the effectiveness of the proposed estimation algorithm at steady state modes of operation. For, respectively, positive and negative rotational speed, Fig. 9(a) and (b) show estimated mechanical speed, actual mechanical speed, current of phase A and estimated position when the mechanical reference speed is 5 rad/s and the rated continuous load torque is applied. Throughout the experimental activity, for oscilloscope visualization purposes, the motor phase currents are measured by means of 200-kHz bandwidth current probes, and all the speed signals are achieved through a 12-b D/A converter.

The experimental tests prove the absolutely negligible error in the speed estimation with respect to the actual speed, which is measured by means of the same 1024 pulses incremental encoder considered for the previous achieved simulations. The Hall effect sensor of stator phase A, which is also shown, confirms both the correctness of the rotor position estimation and the alignment of the phase current with respect the machine phase EMF. The performance of the proposed speed estimation algorithm has been experimentally investigated also at start-up of the AFPM drive and during fast transients, Fig. 10 shows the behavior of the electric drive using the SERF algorithm at reference speed variation from 0 to 6 rad/s. Performances of the drive are very similar to the results observed during simulations: at startup the current in phase A is equal to zero because rotor's magnetic axes position is assumed at the half of the 60° electrical sector identified by means of the Hall sensors' binary code,

when the rotor's magnetic axes enters the first new 60° sector the initial error on the rotor position estimation is reset and each phase current is correctly timed with the respective EMF waveform.

VI. CONCLUSION

In the last few decades, the requirement of removal of unreliable and expensive position sensors has led academic and industrial researchers to investigate and to propose several methods for the sensorless control of PMSM drives. However, low reliability and implementation difficulties of these methods at very low rotational speed modes of operation, preclude their acceptance in direct drive PMSMs devoted to traction applications. In order to overcome the mentioned disadvantages, in this paper the application of a back-EMF-based method in conjunction with very low resolution sensors is proposed to estimate the rotor position of a direct-drive axial-flux permanent-magnet motor to be used for traction purpose of an electric wheelchair. The investigated Speed Estimator is based on the Rotor Frame machine model (SERF) and it has been implemented on a 16-b fixed point DSP. It is proved that SVM techniques, which are based on vector representation of the inverter, divide the plane in six sectors identified by a binary code in the same way the Hall effect sensors provide 60 electrical degree resolution in the rotor position sensing; thus, sector identification from Hall sensors and estimated position from SERF algorithm can be promptly used in SVM logic. Mathematical operations are saved and total execution time of the algorithm is reduced when d - q voltages calculated in the torque and flux loops are directly used in the speed and position estimator, instead of measuring inverter output line-to-line voltages and then transforming them on d - q axis by means of Park and Stanley matrixes. Accurate investigations have been dedicated to startup and near-zero speed operations for the drive, which have been analyzed by means of simulations and experimental tests. Final results show absolutely negligible error in the speed estimation with respect to the actual speed, which is measured by means of an incremental encoder, both in steady-state and in transient operating conditions.

REFERENCES

- [1] H. Kim, M. C. Harke, and R. D. Lorenz, "Sensorless control of interior permanent-magnet machine drives with zero-phaselag position estimation," *IEEE Trans. Ind. Appl.*, vol. 39, no. 6, pp. 1726–1733, Nov./Dec. 2003.
- [2] S. Morimoto, M. Sanada, and Y. Takeda, "Sinusoidal current drive system of permanent magnet synchronous motor with low resolution position sensor," in *Proc. IEEE IAS Annual Meeting*, Oct. 9–13, 1996, vol. 1, pp. 9–14.
- [3] J. Bu, L. Xu, T. Sebastian, and B. Liu, "Near-zero speed performance enhancement of PM synchronous machines assisted by low cost Hall effect sensors," in *Proc. IEEE APEC'98*, Feb. 68–74, 1998, pp. 68–74.
- [4] S. Morimoto, M. Sanada, and Y. Takeda, "High performance current-sensorless drive for PMSM and SynRM with only low resolution position sensor," *IEEE Trans. Ind. Appl.*, vol. 39, no. 3, pp. 792–801, May/Jun. 2003.
- [5] F. G. Capponi, G. De Donato, and L. Del Ferraro, "Brushless AC drive using an axial flux synchronous motor with low resolution position sensors," in *Proc. IEEE 35th Annu. Power Electron. Spec. Conf. (PESC'04)*, Aachen, Germany, 2004, pp. 2287–2292.

- [6] L. Solero, O. Honorati, F. Caricchi, and F. Crescimbin, "Non-conventional three-wheel electric vehicle for urban mobility," *IEEE Trans. Veh. Technol.*, vol. 50, no. 4, pp. 1085–1091, Jul. 2001.
- [7] M. Raganella, A. Di Napoli, F. Crescimbin, A. Lidozzi, and L. Solero, "Design and modeling of controllers in PM drives for wheelchairs," in *Proc. ICEM'04*, 2004, [CD ROM].
- [8] H. W. van der Broeck, H. C. Skudelny, and G. V. Stanke, "Analysis and realization of a pulse width modulator based on voltage space vectors," *IEEE Trans. Ind. Appl.*, vol. IA-24, no. 1, pp. 142–150, Jan./Feb. 1988.
- [9] G. Vitale, A. Di Napoli, F. Crescimbin, A. Lidozzi, and L. Solero, "Combination of SVM techniques for electric drive," in *Proc. SPEEDAM'04*, 2004, [CD ROM].
- [10] J.-W. Choi and S.-K. Sul, "A new compensation strategy reducing voltage/current distortion in PWM VSI systems operating with low output voltages," *IEEE Trans. Ind. Appl.*, vol. 31, no. 5, pp. 1001–1008, Sep./Oct. 1995.
- [11] L. Sang-Bin, T. G. Habetler, R. G. Harley, and D. J. Gritter, "An evaluation of model-based stator resistance estimation for induction motor stator winding temperature monitoring," *IEEE Trans. Energy Conv.*, vol. 17, no. 1, pp. 7–15, Mar. 2002.



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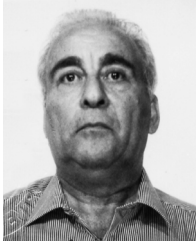


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